

# Onat Gungor, Ph.D.

## Assistant Professor

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Google Scholar

## Research Profile

Trustworthy artificial intelligence for cyber-physical and autonomous systems, with emphasis on **efficient, secure, and reliable AI** in safety-critical environments. Research interests include edge AI, adversarial robustness, multimodal learning, LLM-based agents, cybersecurity, collaborative and autonomous systems, and embodied intelligence.

## Academic Appointments

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| <b>West Virginia University, Department of Computer Science and Electrical Engineering</b><br>Assistant Professor; Director, Trustworthy, Efficient, and Reliable AI Systems (TERA) Lab | 2026–Present |
| <b>University of California San Diego, Computer Science and Engineering</b><br>Postdoctoral Scholar, advised by Prof. Tajana Rosing                                                     | 2024–2026    |
| <b>UC San Diego &amp; San Diego State University, Electrical and Computer Engineering</b><br>Graduate Student Researcher, advised by Prof. Tajana Rosing and Prof. Baris Aksanli.       | 2019–2023    |

## Education

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| <b>Ph.D., Electrical and Computer Engineering</b><br>University of California San Diego & San Diego State University Joint Doctoral Program.<br>Dissertation: <i>Towards Intelligent, Secure, and Efficient Industrial Internet of Things.</i> | 2023 |
| <b>B.S., Computer Science and Engineering</b><br>Ozyegin University. High Honor Degree; graduated at the top of the department.                                                                                                                | 2019 |
| <b>B.S., Industrial Engineering</b><br>Ozyegin University. Valedictorian Award.                                                                                                                                                                | 2018 |

## Peer-Reviewed Publications

### Journal Articles

- J. Chen, **O. Gungor**, Z. Shang, and T. Rosing, “SAGE: Sample-Aware Guarding Engine for Robust Intrusion Detection Against Adversarial Attacks,” *Computer Networks*, 2026.
- C. I. Kocal, **O. Gungor**, T. Rosing, and B. Aksanli, “ReLATE+: Unified Framework for Adversarial Attack Detection, Classification, and Resilient Model Selection in Time-Series Classification,” *Electronics*, 2026.
- S. Nayak, **O. Gungor**, and T. Rosing, “Security in Collaborative Driving: A Survey of Threats, Defenses, and Emerging Trends,” *Electronics*, 2026.
- E. Li, **O. Gungor**, Z. Shang, J. Chen, and T. Rosing, “CITADEL: Continual Anomaly Detection for Enhanced Learning in IoT Intrusion Detection,” *Computer Networks*, 2026.
- O. Gungor**, T. Rosing, and B. Aksanli, “STEWART: Stacking Ensemble for White-Box Adversarial Attacks Towards More Resilient Data-Driven Predictive Maintenance,” *Computers in Industry*, 2022.
- O. Gungor**, T. Rosing, and B. Aksanli, “DOWELL: Diversity-Induced Optimally Weighted Ensemble Learner for Predictive Maintenance of Industrial Internet of Things Devices,” *IEEE Internet of Things Journal*, 2021.
- O. Gungor**, T. Rosing, and B. Aksanli, “RESPIRE++: Robust Indoor Sensor Placement Optimization under Distance Uncertainty,” *IEEE Sensors Journal*, 2021.
- O. Gungor**, B. Aksanli, and R. Aydogan, “Algorithm Selection and Combining Multiple Learners for Residential Energy Prediction,” *Future Generation Computer Systems*, 2019.

### Conference Proceedings

- N. P. Pandey, J. Kong, L. Hu, Q. Zhao, Y. Zhao, **O. Gungor**, H. Zhang, and T. Rosing, “AgentKVShift: Efficient KV Cache Reuse for Agentic Memory Systems,” *Neural Information Processing Systems (NeurIPS)*, 2026.
- N. P. Pandey, J. Park, J. Kong, **O. Gungor**, and T. Rosing, “NgramRL: Accelerating RL Post-Training for Structured Generation via Rollout-Aware N-gram Speculative Decoding,” *Asia and South Pacific Design Automation Conference (ASP-DAC)*, 2027.
- S. Sharma, N. P. Pandey, **O. Gungor**, and T. Rosing, “LIMBO: Lifelong Inference-Time Memory and Budget Optimization for LLM Agents,” *IEEE International Conference on Tools with Artificial Intelligence (ICTAI)*, 2026.
- I. Delgado, H. Gupta, L. Shamoon, B. Khatri, N. Sapre, **O. Gungor**, and T. Rosing, “T-SMART: Mechanism-Level Attribution

- for Tool-Augmented Time-Series Question Answering,” *IEEE ICTAI*, 2026.
5. Y. Tian, Z. Wang, **O. Gungor**, X. Fan, and T. Rosing, “LifeAgentBench: A Multi-dimensional Benchmark and Agent for Personal Health Assistants in Digital Health,” *Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2026.
  6. B. Gupta, **O. Gungor**, and T. Rosing, “G-MARK: Grounded Multi-Agent Reasoning for Cooperative Driving via Knowledge Graphs,” *IEEE International Conference on Machine Learning and Applications (ICMLA)*, 2026.
  7. **O. Gungor**, R. Sood, J. Zhou, and T. Rosing, “EAGER: Edge-Aligned LLM Defense for Robust and Accurate Cybersecurity Question Answering,” *IEEE International Symposium on Networks, Computers and Communications (ISNCC)*, 2026.
  8. W. Lee, P. Mekkoth, Y. Tian, **O. Gungor**, and T. Rosing, “FoodCHA: Multi-Modal LLM Agent for Fine-Grained Food Analysis,” *IEEE ISNCC*, 2026.
  9. G. K. Tokgoz, **O. Gungor**, T. Rosing, and B. Aksanli, “INTARG: Informed Real-Time Adversarial Attack Generation for Time-Series Regression,” *IEEE ISNCC*, 2026.
  10. N. P. Pandey, J. Park, **O. Gungor**, F. Ponzina, and T. Rosing, “QMC: Efficient SLM Edge Inference via Outlier-Aware Quantization and Emergent Memories Co-Design,” *IEEE/ACM ISLPED*, 2026.
  11. J. Chen, A. Deevi, **O. Gungor**, and T. Rosing, “CAN-QA: A Question-Answering Benchmark for Reasoning over In-Vehicle CAN Traffic,” *IEEE International Conference on Computer Communications and Networks (ICCCN)*, 2026.
  12. **O. Gungor**, I. Kale, J. Zhou, and T. Rosing, “LIGHT-HIDS: A Lightweight and Effective Machine Learning-Based Framework for Robust Host Intrusion Detection,” *IEEE ICMLA*, 2025.
  13. **O. Gungor**, R. Sood, H. Wang, and T. Rosing, “AQUA-LLM: Evaluating Accuracy, Quantization, and Adversarial Robustness Trade-offs in LLMs for Cybersecurity Question Answering,” *IEEE ICMLA*, 2025.
  14. Y. Tian, **O. Gungor**, X. Yu, and T. Rosing, “DailyLLM: Context-Aware Activity Log Generation Using Multi-Modal Sensors and LLMs,” *IEEE International Conference on Mobile Ad-Hoc and Smart Systems (MASS)*, 2025.
  15. N. P. Pandey, S. Kulkarni, D. Wang, **O. Gungor**, F. Ponzina, and T. Rosing, “DPQ-HD: Post-Training Compression for Ultra-Low Power Hyperdimensional Computing,” *ACM GLSVLSI*, 2025.
  16. C. I. Kocal, **O. Gungor**, A. Tartz, T. Rosing, and B. Aksanli, “ReLATE: Resilient Learner Selection for Multivariate Time-Series Classification Against Adversarial Attacks,” *IEEE CSR*, 2025.
  17. S. Fuhrman, **O. Gungor**, and T. Rosing, “CND-IDS: Continual Novelty Detection for Intrusion Detection Systems,” *ACM/IEEE DAC*, 2025.
  18. L. Zhang, Q. Zhao, R. Wang, S. Bian, **O. Gungor**, F. Ponzina, and T. Rosing, “ORCA: Offload Rethinking by Cloud Assistance for Efficient Environmental Sound Recognition on LPWAnS,” *ACM SenSys*, 2025.
  19. L. Zhang, F. Ponzina, **O. Gungor**, and T. Rosing, “E-QUARTIC: Energy Efficient Edge Ensemble of Convolutional Neural Networks for Resource-Optimized Learning,” *IEEE/ACM ASP-DAC*, 2025.
  20. **O. Gungor**, A. Rios, P. Mudgal, N. Ahuja, and T. Rosing, “A Robust Framework for Evaluation of Unsupervised Time-Series Anomaly Detection,” *ICPR*, 2024.
  21. F. Asgarinejad, F. Ponzina, **O. Gungor**, T. Rosing, and B. Aksanli, “HDXpose: Harnessing Hyperdimensional Computing’s Explainability for Adversarial Attacks,” *ACM/IEEE ICCAD*, 2024.
  22. **O. Gungor**, T. Rosing, and B. Aksanli, “A2HD: Adaptive Adversarial Training for Hyperdimensional Computing-Based Intrusion Detection Against Adversarial Attacks,” *IEEE CSR*, 2024.
  23. **O. Gungor**, E. Li, Z. Shang, Y. Guo, J. Chen, J. Davis, and T. Rosing, “Rigorous Evaluation of Machine Learning-Based Intrusion Detection Against Adversarial Attacks,” *IEEE CSR*, 2024.
  24. **O. Gungor**, T. Rosing, and B. Aksanli, “ROLDEF: Robust Layered Defense for Intrusion Detection Against Adversarial Attacks,” *DATE*, 2024.
  25. X. Yu, M. Zhou, F. Asgarinejad, **O. Gungor**, B. Aksanli, and T. Rosing, “Private and Secure Edge AI with Hyperdimensional Computing,” *ACM/IEEE DAC*, 2023.
  26. M. Timken, **O. Gungor**, T. Rosing, and B. Aksanli, “Analysis of Machine Learning Algorithms for Cyber Attack Detection in SCADA Power Systems,” *IEEE SmartNets*, 2023.
  27. **O. Gungor**, T. Rosing, and B. Aksanli, “HD-IIoT: Hyperdimensional Computing for Resilient Industrial Internet of Things Analytics,” *DATE*, 2023.
  28. **O. Gungor**, T. Rosing, and B. Aksanli, “DENSE-DEFENSE: Diversity Promoting Ensemble Adversarial Training Towards Effective Defense,” *IEEE SENSORS*, 2022.
  29. **O. Gungor**, T. Rosing, and B. Aksanli, “CAHEROS: Constraint-Aware Heuristic Approach for Robust Sensor Placement,” *IEEE SENSORS*, 2021.
  30. **O. Gungor**, T. Rosing, and B. Aksanli, “ENFES: Ensemble Few-Shot Learning for Intelligent Fault Diagnosis with Limited Data,” *IEEE SENSORS*, 2021.
  31. **O. Gungor**, T. Rosing, and B. Aksanli, “OPELRUL: Optimally Weighted Ensemble Learner for Remaining Useful Life Prediction,” *IEEE ICPHM*, 2021.
  32. **O. Gungor**, J. Garnier, T. Rosing, and B. Aksanli, “LENARD: Lightweight Ensemble Learner for Medium-Term Electricity Consumption Prediction,” *IEEE SmartGridComm*, 2020.
  33. **O. Gungor**, T. Rosing, and B. Aksanli, “RESPIRE: Robust Sensor Placement Optimization in Probabilistic Environments,” *IEEE SENSORS*, 2020. **Best Paper Nominee.**

## Workshop Papers and Extended Abstracts

1. J. Chen, **O. Gungor**, Z. Shang, E. Li, and T. Rosing, “DYNAMITE: Dynamic Defense Selection for Enhancing Machine Learning-Based Intrusion Detection Against Adversarial Attacks,” *IEEE/ACM Workshop on the Internet of Safe Things*, 2025.

### Best Paper Award.

2. **O. Gungor**, A. Rios, N. Ahuja, and T. Rosing, "TS-OOD: An Evaluation Framework for Time-Series Out-of-Distribution Detection and Prospective Directions for Progress," *AAAI Workshop on AI for Time Series Analysis*, 2025.
3. E. Li, Z. Shang, **O. Gungor**, and T. Rosing, "SAFE: Self-Supervised Anomaly Detection Framework for Intrusion Detection," *AAAI Workshop on AI for Cyber Security*, 2025.
4. **O. Gungor**, T. Rosing, and B. Aksanli, "Adversarial-HD: Hyperdimensional Computing Adversarial Attack Design for Secure Industrial Internet of Things," *IEEE/ACM Workshop on the Internet of Safe Things*, 2023. **Best Paper Runner-Up**.
5. **O. Gungor**, U. Cakan, R. Aydogan, and P. Ozturk, "Effect of Awareness of Other Side's Gain on Negotiation Outcome, Emotion, Argument and Bidding Behavior," *International Workshop on Agent-Based Complex Automated Negotiation*, 2019. **Best Student Paper**.
6. Y. Tian, **O. Gungor**, X. Yu, and T. Rosing, "Fine-Grained Contextualized Activity Logs Generation Based on Multi-Modal Sensor Data and LLM," Poster Abstract, *ACM SenSys*, 2025.

## Honors and Awards

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- **Best Paper Award**, IEEE/ACM Workshop on the Internet of Safe Things, 2025 (DYNAMITE).
- **Best Paper Runner-Up**, IEEE/ACM Workshop on the Internet of Safe Things, 2023 (Adversarial-HD).
- San Diego State University **University Graduate Fellowship**, 2021.
- **Best Paper Nominee**, IEEE SENSORS, 2020 (RESPIRE).
- **Best Student Paper**, International Workshop on Agent-Based Complex Automated Negotiation (ACAN), 2019.

## Selected Research Collaborations

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- **Intel Labs**: time-series anomaly detection, phase detection, and out-of-distribution detection.
- **Google**: multimodal and LLM-powered personal health agent systems.

## Teaching Experience

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- **Instructor, West Virginia University**: CPE 442 – Introduction to Digital Computer Architecture (Fall 2026).
- **Instructor, UC San Diego**: CSE 15L – Software Tools and Techniques (Spring 2022, Fall 2023); CSE 140 – Components and Design Techniques for Digital Systems (Summer 2022).
- **Teaching Assistant**: SDSU COMPE 375 – Embedded Systems Programming (Summer 2021, Summer 2022); UCSD CSE 140 – Components and Design Techniques for Digital Systems (Winter 2022).

## Student Mentoring and Advising

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Mentored **40+** students since **2020** across Ph.D., M.S., and undergraduate research. Selected mentees include:

- Nilesh Pandey (Ph.D.) – efficient and scalable edge AI; publications at GLSVLSI 2025, ISLPED 2026, ASP-DAC 2027.
- Ye Tian (Ph.D.) – LLMs and physical-world understanding; publications at SenSys 2025, MASS 2025, EMNLP 2026.
- Le Zhang (M.S.) – efficient edge AI and edge-cloud collaboration; ASP-DAC 2025 and SenSys 2025.
- Ipek Kocal (M.S.) – resilient time-series classification under adversarial attacks; CSR 2025 and Electronics 2026.
- Jing Chen, Elvin Li, Zhengli Shang, Ishaan Kale, Jiasheng Zhou, Roshan Sood, and Harold Wang (undergraduate researchers) – multiple peer-reviewed publications spanning adversarial ML, intrusion detection, and LLM security.

## Selected Invited Talks

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- "Collaborative Intelligence for Securing the IoT: Robust, Adaptive, and Secure AI-Driven Intrusion Detection Systems," SRC JUMP 2.0 CoCoSys Theme 4 Meeting, May 2025.
- "AI-Driven Cybersecurity: Advancing Intrusion Detection, LLM Security and Time-Series Anomaly Detection," Cisco Foundation AI Group, April 2025.

## Academic Service and Outreach

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**Journal Reviewer**: IEEE Internet of Things Journal; IEEE Transactions on Information Forensics and Security; IEEE Transactions on Dependable and Secure Computing; IEEE Transactions on Industrial Informatics; IEEE Transactions on Industrial Electronics; IEEE Transactions on Network and Service Management.

**Conference Reviewer**: DATE 2027; ASPLOS 2025; DATE 2025; ECAI 2024–2025; MICRO 2024; CODES+ISSS 2023; DAC 2023; IEEE SENSORS 2022.

**Research Outreach**: ENLACE bi-national summer research program; UC San Diego Early Research Scholars Program.